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Decision Process Engineering

Data Analysis Presentation

Multiple Linear Regression Analysis: Student Intervention Dataset

Quantifying Intervention Methods' Impact on Student
Math Scores Performance

A Brief Introduction

- ❑ Gareth Fernes
- ❑ Bachelor of Architecture, Florida Atlantic University
- ❑ Currently pursuing a Master's in Data Analytics and Decision Process Engineering at Western Governors University.
- ❑ My interests involve analyzing financial data and technology-related stocks.
- ❑ My interest in this topic is because early hands-on intervention in kids' education, especially in math, reading, writing, and speaking, may benefit them early and make them more confident to deal with educational challenges later in school.

Statement of Problem

How can we help students by prioritizing the most effective intervention methods to help them improve their math scores?

Research Question

Can a multiple regression model be constructed on a student intervention dataset to statistically predict math score improvement?

Hypothesis

Null Hypothesis: A multiple regression model constructed on a student success dataset statistically will not achieve a predictive accuracy exceeding 70% ($R\text{-Squared} \leq 0.70$) in predicting math score improvement.

Alternate Hypothesis: A multiple regression model constructed from a student success dataset will statistically achieve a predictive accuracy exceeding 70% ($R\text{-Squared} > 0.70$) in predicting math score improvement.

What are we trying to solve?

Schools use different intervention methods to help students improve their math scores. But which method works best?

This study uses data analytics to answer that question using student performance metrics. Four teaching methods were compared.

- ❑ Tutoring: One-on-one or small group instruction.
- ❑ Blended Learning: A mix of online and in-person teaching.
- ❑ Flipped Classroom: Students watch lessons at home, practice in class.
- ❑ Control Group: Traditional classroom instruction with no special intervention.

Data Preparation

- ❑ First, we had to download the data and verify that it was suitable for regression analysis.
- ❑ The dataset is a .csv file that tracks student metrics.
- ❑ It consists of 16 columns and 10,000 rows downloaded from Kaggle (www.kaggle.com), a public data platform.
- ❑ All data were handled in accordance with standard data privacy practices, and no individual student records were shared or published.
- ❑ RStudio was used as the primary tool for all stages of data extraction and preparation. It is open source and is specifically designed for statistical computing and data analysis.
- ❑ It supports reproducible research through R Markdown. Research can be shared and collaborated on.
- ❑ Packages used during data analysis included *tidyverse* for data manipulation, *ggplot2* for data visualization and exploratory analysis, *car* for VIF-based multicollinearity diagnostics, and *caret* for model evaluation, cross-validation, and R-Squared assessment.

Data Preparation (continued)

- ❑ A sample of the data used in the study.
- ❑ The highlighted 7 of 16 columns were removed because they were not included in the study.
- ❑ A new column *math improvement* was created by calculating the difference between each student's post-intervention math score and their baseline math score. A total of 10 variables were used in the study.

COLUMN NAME	TYPE	SAMPLE DATA
student_id	Continuous	93207890110
school_id	Continuous	110
grade_level	Continuous	9
intervention	Categorical	blended
ses_index	Continuous	1.379
attendance_rate	Continuous	0.996
study_hours_wk	Continuous	5.93
teacher_experience_years	Continuous	6.7
class_size	Continuous	32
baseline_math	Continuous	78.9
baseline_reading	Continuous	82.3
baseline_motivation_1_10	Continuous	7.1
post_math	Continuous	88.9
post_reading	Continuous	95.2
test_anxiety_1_10	Continuous	3.8
passed	Continuous	1

Data Preparation (continued)

- ☐ Check for missing data. There was no missing data.
- ☐ No data was imputed or changed for this study.
- ☐ Removed irrelevant columns for this study.
- ☐ All outliers were retained for the research.

Data Analysis

- ❑ Prior to building the regression model, exploratory data analysis was conducted
- ❑ To visualize the relationship between intervention type and math score improvement.
- ❑ Tested and plotted the data (Q-Q Plot, Histogram, Box Plot, VIF Test), to check for anomalies.
- ❑ Multiple linear regression was selected because it simultaneously evaluates the relationships among multiple independent variables and a single continuous dependent variable (*math score improvement*).
- ❑ In this study, a technique called Bidirectional Stepwise Regression was used, which automatically removes variables that do not contribute meaningfully to the model.
- ❑ This left us with 7 key predictors out of the original 9, making our model cleaner and more efficient (*ses_index and test_anxiety were removed*).
- ❑ An 80/20 train-test split to build the model.

Data Analysis (continued)

- ❑ Average values of the 10 variables included in the study.
- ❑ The average improvement in math scores was 10.87 points.
- ❑ The average student attendance rate was 91.3%.
- ❑ The average student studies an average of 4.58 hours per week.
- ❑ The average teaching experience was 9.06 years.
- ❑ The average class size was 26 students.
- ❑ The average baseline math score was 70.03.

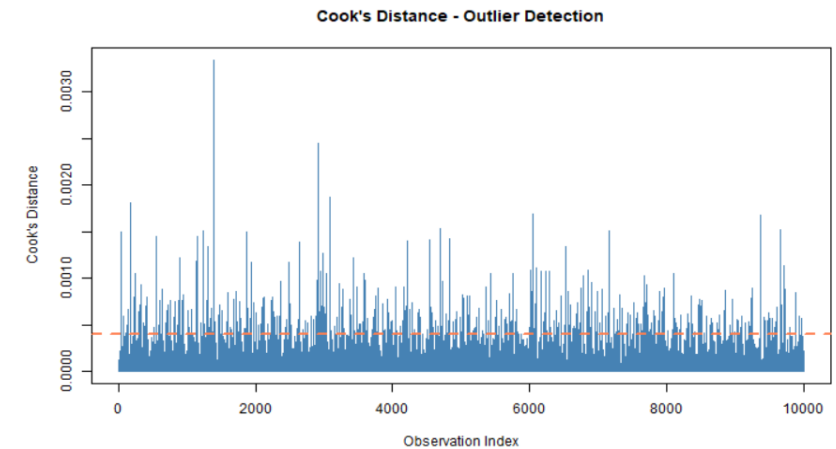
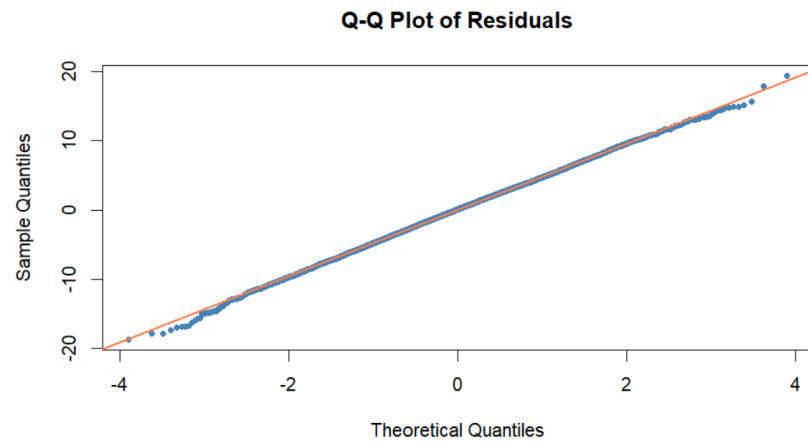
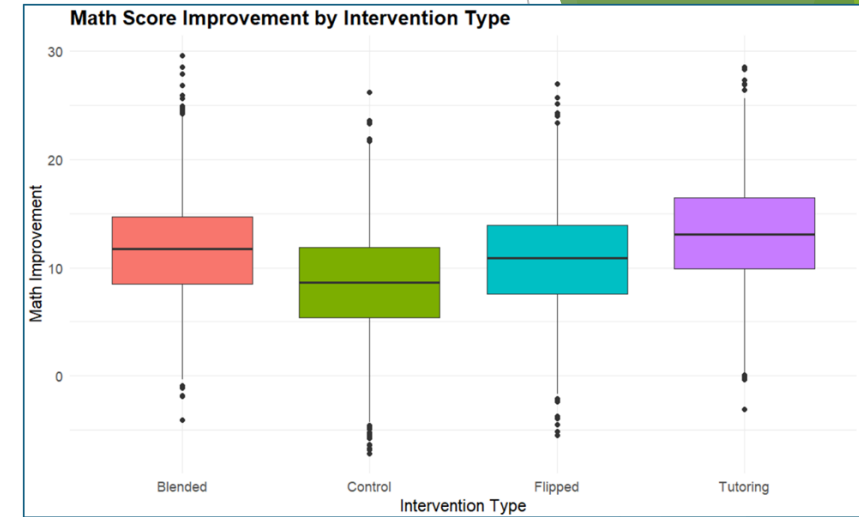
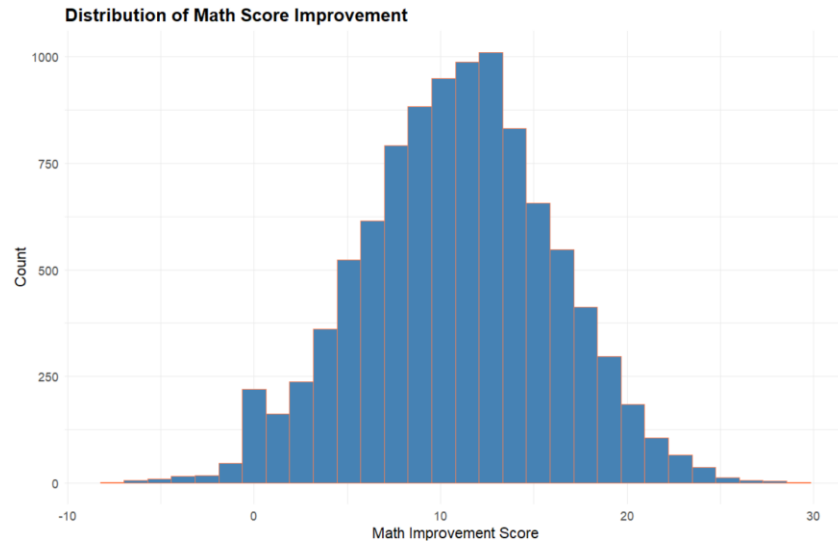
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## DATA ANALYSIS
### Summary of the Data
{r summary_data}
summary(student_math)
```

intervention	ses_index	attendance_rate	study_hours_wk	teacher_experience_years	
class_size					
Min. :1.000	Min. : -2.500000	Min. :0.6670	Min. : 0.020	Min. : 0.000	Min.
:10.00					
1st Qu.:2.000	1st Qu.: -0.678250	1st Qu.:0.8730	1st Qu.: 2.250	1st Qu.: 5.600	1st
Qu.:22.00					
Median :2.000	Median : 0.001000	Median :0.9160	Median : 3.910	Median : 9.000	Median
:26.00					
Mean :2.512	Mean : -0.002791	Mean :0.9129	Mean : 4.576	Mean : 9.057	Mean
:26.06					
3rd Qu.:4.000	3rd Qu.: 0.671000	3rd Qu.:0.9570	3rd Qu.: 6.140	3rd Qu.:12.400	3rd
Qu.:30.00					
Max. :4.000	Max. : 2.500000	Max. :1.0000	Max. :22.850	Max. :28.700	Max.
:40.00					
baseline_math	baseline_motivation_1_10	test_anxiety_1_10	math_improvement		
Min. : 20.00	Min. : 1.000	Min. : 1.000	Min. : -7.20		
1st Qu.: 61.00	1st Qu.: 5.300	1st Qu.: 3.700	1st Qu.: 7.40		
Median : 70.00	Median : 6.500	Median : 4.800	Median :11.00		
Mean : 70.03	Mean : 6.456	Mean : 4.764	Mean :10.87		
3rd Qu.: 79.20	3rd Qu.: 7.600	3rd Qu.: 5.900	3rd Qu.:14.30		
Max. :100.00	Max. :10.000	Max. :10.000	Max. :29.60		

Data Analysis (continued)

- ❑ The model trained on 80% of the data (8,001 records) achieved a training R-Squared of 13.84%, while the test R-Squared on the remaining 20% (1,999 records) was 13.33%.
- ❑ The minimal difference of 0.51% between the training and test R-Squared values confirms that the model generalizes well with new data.
- ❑ The final stepwise model produced an R-Squared of 0.1375 (13.75%) and an Adjusted R-Squared of 0.1369.
- ❑ The cross-validated R-Squared of 13.72% was highly consistent with the stepwise model R-Squared of 13.75%, confirming that the model is stable and does not overfit.

Data Analysis (continued)



Visual check for normality and outliers in the data for the regression analysis

Key Findings

The analysis revealed six significant factors that impact student math improvement. In order from top to bottom, the impact of each variable on math score improvement.

- ❑ Intervention Type = FIRST.
- ❑ Attendance = SECOND.
- ❑ Study time = THIRD.
- ❑ Teachers Experience = FOURTH.
- ❑ Class Size= FIFTH.
- ❑ Baseline Math Score = SIXTH.

Outcome of the Research

- ☐ The hypothesis for this research is that our model achieves 70% predictive accuracy.
- ☐ Our model achieved 13.75%, which means we did not meet that threshold.
- ☐ However, this does not mean the model failed. It means that while our identified factors are statistically significant, other unmeasured factors, such as parental involvement, student mental health, and home environment, also play major roles in math improvement.

Which Teaching Method Worked Best

When comparing all four teaching methods.

- ☐ Tutoring produced the greatest improvement in math scores.
- ☐ Followed by blended learning and the flipped classroom.
- ☐ The control group's traditional instruction showed the lowest improvement.

Limitations

- ❑ A disadvantage of using secondary data is the lack of control over how the data was originally collected.
- ❑ The dataset does not provide information on how students were assigned to intervention groups, raising concerns about selection bias.
- ❑ The dataset excludes qualitative factors such as parental involvement, student mental health, and home environment, which are well-documented influences on academic performance.
- ❑ If students were not randomly assigned, pre-existing differences between groups, such as prior academic achievement or socioeconomic background, could influence the outcomes independently of the intervention method.

Limitations (continued)

- ❑ A disadvantage of multiple linear regression is its assumption of linearity.
- ❑ The technique assumes a straight-line relationship between each predictor and the dependent variable.
- ❑ If the true relationship between predictors and math score improvement is non-linear, the model will fail to capture these patterns, resulting in reduced predictive accuracy.
- ❑ The low R-Squared of 13.75% observed in this study suggests that linear regression may not fully capture the complexity of the relationships in the data.

Benefits of the Study

- ❑ The analysis confirmed that a statistically significant multiple linear regression model can be constructed, with the final stepwise regression model producing an F-statistic of 227.6 ($p < 2.2e-16$), indicating the model is highly significant overall.
- ❑ The consistency of R-Squared values across all evaluation methods, full model (13.76%), stepwise model (13.75%), cross-validation (13.72%), and train-test split test data (13.33%), confirms that the model is stable, does not overfit, and generalizes reliably to new data.
- ❑ Based on this analysis, schools and educational administrators should prioritize tutoring as the primary academic intervention for students seeking to improve their math scores (Beta = 0.708).
- ❑ Attendance rate was the second strongest predictor of math improvement (Beta = 5.307).
- ❑ Students who studied more hours per week showed greater math improvement (Beta = 0.309).

Conclusion and Proposed Actions

This study provides schools and administrators with data-driven evidence that intervention methods matter. Schools looking to maximize student math performance should prioritize programs.

- ☐ Tutoring shows the strongest impact on math score improvement.
- ☐ Improve attendance policies (the next strongest predictor of math improvement).
- ☐ Reduce class sizes where possible.
- ☐ Students who studied more hours per week showed greater math improvement
- ☐ Prioritizing experienced teachers.

Future Research

Future studies should incorporate additional variables to build a more comprehensive and accurate predictive model. Additionally, applying non-linear modeling techniques, such as Random Forests or Gradient Boosting, to the same dataset may capture complex relationships that linear regression cannot detect and yield significantly higher predictive accuracy.

- ☐ Parental involvement.
- ☐ Learning disabilities.
- ☐ Socioeconomic conditions.

Sources

Arrington, Edith G. (2024). Academic Interventions Retrieved March 13th, 2026

<https://www.ebsco.com/research-starters/education/academic-interventions>

R for Data Science (Wickham, H. 2016). Retrieved February 1st, 2026 <https://r4ds.hadley.nz/>

Thulin, M. (2026). Modern Statistics with R

<https://www.modernstatisticswithr.com/>

Aissaoui, Ouafae & Madani, Yasser & Oughdir, Lahcen & Dakkak, Ahmed & el Alloui, Youssef. (2020). A Multiple Linear Regression-Based Approach to Predict Student Performance. Retrieved March 13th, 2026

[10.1007/978-3-030-36653-7_2](https://doi.org/10.1007/978-3-030-36653-7_2).

Dr. Sewell, W (2026). Education_Intervention_Dataset Retrieved February 1st, 2026

<https://www.kaggle.com/datasets/williamsewell/education-intervention-dataset>